

PRACTICE SHEET-1

1. What sum of money must be given as simple interest for six months at 4% per annum in order to earn 150 interest?
 (1) 5000 (2) 7500
 (3) 10000 (4) 15000
2. A sum of 1600 gives a simple interest of 252 in 2 years and 3 months. The rate of interest per annum is:
 (1) 5.5 % (2) 8%
 (3) 7% (4) 6%
3. A sum of money lent at simple interest amounts to 880 in 2 years and to 920 in 3 years. The sum of money (in rupees) is
 (1) 700 (2) 760
 (3) 784 (4) 800
4. At some rate of simple interest, A lent 6,000 to B for 2 years and 1,500 to C for 4 years and received 9,00 as interest from both of them together. The rate of interest per annum was
 (1) 5% (2) 6%
 (3) 8% (4) 10%
5. A lent 5000 to B for 2 years and 3000 to C for 4 years on simple interest at the same rate of interest and received 2200 in all from both as interest. The rate of interest per annum is
 (1) 7% (2) 5%
 (3) 7.125% (4) 10%
6. What sum of money will amount to 520 in 5 years and to 568 in 7 years at simple interest?
 (1) 400 (2) 120
 (3) 510 (4) 220
7. 500 was invested at 12% per annum simple interest and a certain sum of money invested at 10% per annum simple interest. If the sum of the interest on both the sum after 4 years is 480, the latter sum of money is:
 (1) 450 (2) 750
 (3) 600 (4) 550
8. A money lender finds that due to fall in the annual rate of interest 8% to 7.75 %, his yearly income diminishes by 61.50. His capital is
 (1) 22400 (2) 23800
 (3) 24600 (4) 26000
9. A lends 2500 to B and a certain sum to C at the same time at 7% annual simple interest. If after 4 years, A altogether receives 1120 as interest from B and C, the sum lent to C is
 (1) 700 (2) 6500
 (3) 4000 (4) 1500
10. A certain sum of money amounts to 756 in 2 years and to 873 in 3 and half years at a certain rate of simple interest. The rate of interest per annum is
 (1) 10% (2) 11%
 (3) 12% (4) 13%
11. What sum will amount to 7000 in 5 years at $3\frac{1}{3}$ % simple interest?
 (1) 6300 (2) 6500
 (3) 6000 (4) 5000
12. A man took a loan from a bank at the rate of 12% per annum at simple interest. After 3 years he had to pay 5,400 as interest only for the period. The principal amount borrowed by him was:
 (1) 2,000 (2) 10,000
 (3) 20,000 (4) 15,000
13. A sum of money at simple interest amounts to 1,012 in $2\frac{1}{2}$ years and to 1,067.20 in 4 years. The rate of interest per annum is :
 (1) 2.5% (2) 3%
 (3) 4% (4) 5%
14. A sum of money lent out at simple interest amounts to 720 after 2 years and to 1020 after a further period of 5 years. The sum is :
 (1) 500 (2) 600
 (3) 700 (4) 710
15. The sum of money, that will give 1 as interest per day at the rate of 5% per annum simple interest is
 (1) 3650 (2) 36500
 (3) 730 (4) 7300

- 16.** A sum of money becomes $\frac{7}{6}$ of itself in 3 years at a certain rate of simple interest. The rate per annum is:
 (1) $5\frac{5}{9}\%$ (2) $6\frac{5}{9}\%$
 (3) 18% (4) 25%
- 17.** A sum of money becomes $\frac{41}{40}$ of itself in $\frac{1}{4}$ years at a certain rate of simple interest. The rate of interest per annum is
 (1) 10% (2) 1%
 (3) 2.5% (4) 5%
- 18.** A certain sum of money becomes three times of itself in 20 years at simple interest. In how many years does it become double of itself at the same rate of simple interest?
 (1) 8 years (2) 10 years
 (3) 12 years (4) 14 years
- 19.** At what rate per cent per annum will the simple interest on a sum of money be $\frac{2}{5}$ of the amount in 10 years?
 (1) 4% (2) 6%
 (3) $5\frac{2}{3}\%$ (4) $6\frac{2}{3}\%$
- 20.** 6,000 become 7,200 in 4 years at a certain rate of simple interest. If the rate becomes 1.5 times of itself, the amount of the same principal in 5 years will be
 (1) 8,000 (2) 8,250
 (3) 9,250 (4) 9,000
- 21.** A sum of money at simple interest trebles itself in 15 years. It will become 5 times of itself in
 (1) 40 years (2) 36 years
 (3) 30 years (4) 25 years
- 22.** If a sum of money at simple interest doubles in 12 years, the rate of interest per annum is
 (1) $16\frac{2}{3}\%$ (2) 7.5%
 (3) $8\frac{1}{3}\%$ (4) 10%
- 23.** At what rate of simple interest per annum will a sum become $\frac{7}{4}$ of itself in 4 years?
 (1) 18% (2) $18\frac{1}{4}\%$
 (3) $18\frac{3}{4}\%$ (4) $18\frac{1}{2}\%$
- 24.** A sum of money at a certain rate per annum of simple interest doubles in the 5 years and at a different rate becomes three times in 12 years. The lower rate of interest per annum is
 (1) 15% (2) 20%
 (3) $15\frac{3}{4}\%$ (4) $16\frac{2}{3}\%$
- 25.** In how many years will a sum of money double itself at $6\frac{1}{4}\%$ simple interest per annum?
 (1) 24 years (2) 20 years
 (3) 16 years (4) 12 years
- 26.** At a certain rate of simple interest, a certain sum of money becomes double of itself in 10 years. It will become treble of itself in
 (1) 15 years (2) 18 years
 (3) 20 years (4) 30 years
- 27.** In how much time, will a sum of money become double of itself at 15% per annum simple interest?
 (1) $6\frac{1}{4}$ years (2) $6\frac{1}{2}$ years
 (3) $6\frac{1}{3}$ (4) $6\frac{2}{3}$ years
- 28.** In how many years will a sum of money double itself at 12% per annum?
 (1) 8 yrs. 6 months (2) 6 yrs. 9 months
 (3) 8 yrs. 4 months (4) 7 yrs. 6 months
- 29.** A sum amounts to double in 8 years by simple interest. Then the rate of simple interest per annum is
 (1) 10% (2) 12.5%
 (3) 15% (4) 20%

30. A sum doubles itself in 16 years, then in how many years will it triple itself; rate of interest being simple
- (1) 25 years (2) 24 years
(3) 48 years (4) 32 years

ANSWER KEY

1.	(2)	2.	(3)	3.	(4)	4.	(1)	5.	(4)	6.	(1)	7.	(3)	8.	(4)	9.	(4)	10.	(4)
11.	(3)	12.	(4)	13.	(3)	14.	(2)	15.	(4)	16.	(1)	17.	(1)	18.	(2)	19.	(4)	20.	(2)
21.	(3)	22.	(3)	23.	(3)	24.	(4)	25.	(3)	26.	(3)	27.	(4)	28.	(3)	29.	(2)	30.	(4)

PRACTICE SHEET-2

- In what time will the simple interest be $\frac{2}{5}$ of the principal at 8 per cent per annum?
(1) 8 years (2) 7 years
(3) 5 years (4) 6 years
- The simple interest on a sum after 4 years is $\frac{1}{5}$ of the sum. The rate of interest per annum is:
(1) 4% (2) 5%
(3) 6% (4) 8%
- Simple interest on a certain sum for 6 years is $\frac{9}{25}$ of the sum. The rate of interest is:
(1) 6% (2) $6\frac{1}{2}\%$
(3) 8% (4) $8\frac{1}{2}\%$
- The simple interest on a sum for 5 years is one fourth of the sum. The rate of interest per annum is
(1) 5% (2) 6%
(3) 4% (4) 8%
- On a certain sum, the simple interest at the end of $6\frac{1}{4}$ years becomes $\frac{3}{8}$ of the sum. The rate of interest is
(1) 5% (2) 6%
(3) 7% (4) 8%
- The simple interest on a certain sum at 5% per annum for 3 years and 4 years differ by 42. The sum is :
(1) 210 (2) 280
(3) 750 (4) 840
- The difference between the simple interest received from two different sources on 1500 for 3 years is 13.50. The difference between their rates of interest is:
(1) 0.1% (2) 0.2%
(3) 0.3% (4) 0.4%
- The simple interest on a sum of money is $\frac{4}{9}$ of the principal and the number of years is equal to the rate percent per annum. The rate per annum is:
(1) 5% (2) $6\frac{2}{3}\%$
(3) 6% (4) $7\frac{1}{5}\%$
- The simple interest on a certain sum for 8 months at 4% per annum is 129 less than the simple interest on the same sum for 15 months at 5% per annum. The sum is :
(1) 2,580 (2) 2400
(3) 2529 (4) 3600
- Mohan lent some amount of money at 9% simple interest and an equal amount of money at 10% simple interest each for two years. If his total interest was Rs. 760, what amount was lent in each case?
(1) 1700 (2) 1800
(3) 1900 (4) 2000

- 11.** A sum of 1550 was lent partly at 5% and partly at 8% simple interest. The total interest received after 3 years is 300. The ratio of money lent at 5% to that at 8% is :
 (1) 5 : 8 (2) 8 : 5
 (3) 31 : 6 (4) 16 : 15
- 12.** A person lent 5,000 partly at the rate of 4 per cent and partly at the rate of 5 per cent per annum simple interest. The total interest after 2 years is 440. To find the sum of money lent at each of the above rates, 5,000 is to be divided in the ratio :
 (1) 4 : 5 (2) 3 : 2
 (3) 5 : 4 (4) 2 : 3
- 13.** A person borrows some money for 5 years and loan amount: total interest amount is 5 : 2. The ratio of loan amount: interest rate is equal to :
 (1) 2 : 25 (2) 2 : 1
 (3) 5 : 2 (4) 25 : 2
- 14.** A person invests money in three different schemes for 6 years, 10 years and 12 years at 10 per cent, 12 per cent and 15 per cent simple interest respectively. At the completion of each scheme, he gets the same interest. The ratio of his investment is
 (1) 6 : 3 : 2 (2) 2 : 3 : 4
 (3) 3 : 4 : 6 (4) 3 : 4 : 2
- 15.** With a given rate of simple interest, the ratio of principal and amount for a certain period of time is 4 : 5. After 3 years, with the same rate of interest, the ratio of the principal and amount becomes 5 : 7. The rate of interest is
 (1) 4% (2) 6%
 (3) 5% (4) 7%
- 16.** A sum of 400 amounts to 480 in 4 years. What will it amount to if the rate of interest is increased by 2%?
 (1) 484 (2) 560
 (3) 512 (4) None of these
- 17.** A man loses 55.50 yearly when the annual rate of interest falls from 11.5% to 10%. His capital (in rupees) is
 (1) 3700 (2) 7400
 (3) 8325 (4) 11100
- 18.** If the annual rate of simple interest increases from 10% to $12\frac{1}{2}\%$, a man's yearly income increases by 1250. His principal (in rupees) is
 (1) 50,000 (2) 45,000
 (3) 60,000 (4) 65,000
- 19.** A sum was invested on simple interest at a certain rate for 2 years. Had it been put at 3% higher rate, it would have fetched 72 more. The sum is
 (1) 1,200 (2) 1,500
 (3) 1,600 (4) 1,800
- 20.** A sum of money was lent at simple interest at a certain rate for 3 years. Had it been lent at 2.5% per annum higher rate, it would have fetched 540 more. The money lent was:
 (1) 6400 (2) 6472
 (3) 6840 (4) 7200
- 21.** A sum of money was invested at a certain rate of simple interest for 2 years. Had it been invested at 1% higher rate, it would have fetched 24 more interest. The sum of money is :
 (1) 1200 (2) 1050
 (3) 1000 (4) 9600
- 22.** A person who pays income tax at the rate of 4 paise per rupee, find that a fall of interest rate from 4% to 3.75% diminishes his net yearly income by 48. What is his capital ?
 (1) 24,000 (2) 25,000
 (3) 20,000 (4) 18,000
- 23.** A sum was lent at simple interest at a certain rate for 2 years. Had it been lent at 3% higher rate, it would have fetched 300 more. The original sum of money was
 (1) 5000 (2) 6000
 (3) 7000 (4) 4000
- 24.** A sum of 2,400 amounts to 3,264 in 4 years at a certain rate of simple interest. If the rate of interest is increased by 1%, the same sum in the same time would amount to
 (1) 3,288 (2) 3,312
 (3) 3,340 (4) 3,360
- 25.** 800 amounts to 920 in 3 years at simple interest. If the interest rate is increased by 3%, it would amount to
 (1) 1,056 (2) 1,112
 (3) 1,182 (4) 992

- 26.** A sum of 10,000 is lent partly at 8% and remaining at 10% per annum. If the yearly interest on the average is 9.2%, the two parts are :
 (1) 4000, 6000 (2) 4500, 5500
 (3) 5000, 5000 (4) 5500, 4500
- 27.** A sum of 1000 is lent out partly at 8% and the remaining at 10% per annum. If the yearly income on the average is 9.2%, the two parts respectively are
 (1) 400, 600 (2) 450, 550
 (3) 500, 500 (4) 550, 450
- 28.** An old article is available for 12,000 at cash payment or is available for 7,000 cash payment and a monthly installment of 630 for 8 months. The rate per cent per annum is
 (1) 2.1 per cent (2) 3 per cent
 (3) 3.25 per cent (4) 3.3 per cent
- 29.** The effective annual rate of interest, corresponding to a nominal rate of 6% per annum payable half yearly, is :
 (1) 6.06% (2) 6.07%
 (3) 6.08% (4) 6.09%
- 30.** A person lends 40% of his sum of money at 15% per annum, 50% of rest at 10% per annum and the rest at 18% per annum rate of interest. What would be the annual rate of interest, if the interest is calculated on the whole sum?
 (1) 13.4% (2) 14.33%
 (3) 14.4% (4) 13.33%

ANSWER KEY

1.	(3)	2.	(2)	3.	(1)	4.	(1)	5.	(2)	6.	(4)	7.	(3)	8.	(4)	9.	(4)	10.	(4)
11.	(4)	12.	(2)	13.	(4)	14.	(1)	15.	(3)	16.	(3)	17.	(1)	18.	(1)	19.	(1)	20.	(4)
21.	(1)	22.	(3)	23.	(1)	24.	(4)	25.	(4)	26.	(1)	27.	(1)	28.	(1)	29.	(4)	30.	(3)

PRACTICE SHEET- 3

- 1.** At what percent per annum will 3000/- amounts to 3993/- in 3 years if the interest is compounded annually?
 (1) 9% (2) 10%
 (3) 11% (4) 13%
- 2.** The compound interest on 10,000 in 2 years at 4% per annum, the interest being compounded half-yearly, is :
 (1) 636.80 (2) 824.32
 (3) 912.86 (4) 828.82
- 3.** In how many years will 2,000 amounts to 2,420 at 10% per annum compound interest?
 (1) 3 years (2) $2\frac{1}{2}$ years
 (3) 2 years (4) $1\frac{1}{2}$ years
- 4.** In what time will 1000 becomes 1331 at 10% per annum compounded annually?
 (1) 3 years (2) $2\frac{1}{2}$ years
- (3) 2 years (4) $3\frac{1}{2}$ years
- 5.** The principal, who will amount to 270.40 in 2 years at the rate of 4% per annum compound interest, is
 (1) 200 (2) 225
 (3) 250 (4) 220
- 6.** A sum of money on compound interest amounts to 10648 in 3 years and 9680 in 2 years. The rate of interest per annum is
 (1) 5% (2) 10%
 (3) 15% (4) 20%
- 7.** At what rate per cent per annum will 2304 amount to 2500 in 2 years at compound interest ?
 (1) $4\frac{1}{2}\%$ (2) $4\frac{1}{5}\%$
 (3) $4\frac{1}{6}\%$ (4) $4\frac{1}{3}\%$

- 8.** A sum becomes 1,352 in 2 years at 4% per annum compound interest. The sum is
 (1) 1,225 (2) 1,270
 (3) 1,245 (4) 1,250
- 9.** The compound interest on 16,000 for 9 months at 20% per annum, interest being compounded quarterly, is
 (1) 2,520 (2) 2,524
 (3) 2,522 (4) 2,518
- 10.** If the rate of interest be 4% per annum for first year, 5% per annum for second year and 6% per annum for third year, then the compound interest of 10,000 for 3 years will be
 (1) 1,600 (2) 1,625.80
 (3) 1,575.20 (4) 2,000
- 11.** The compound interest on 2000 in 2 years if the rate of interest is 4% per annum for the first year and 3% per annum for the second year, will be
 (1) 142.40 (2) 140.40
 (3) 141.40 (4) 143.40
- 12.** At what rate per annum will 32000 yield a compound interest of 5044 in 9 months interest being compounded quarterly?
 (1) 20% (2) 32%
 (3) 50% (4) 80%
- 13.** The compound interest on 8,000 at 15% per annum for 2 years 4 months, compounded annually is:
 (1) 2980 (2) 3091
 (3) 3109 (4) 3100
- 14.** In what time will 10,000 amount to 13310 at 20% per annum compounded half yearly?
 (1) $1\frac{1}{2}$ years (2) 2 years
 (3) $2\frac{1}{2}$ years (4) 3 years
- 15.** A certain sum of money yields 1261 as compound interest for 3 years at 5% per annum. The sum is
 (1) 9000 (2) 8400
 (3) 7500 (4) 8000
- 16.** A certain sum, invested at 4% per annum compound interest, compounded half yearly, amounts to 7,803 at the end of one year. The sum is
 (1) 7,000 (2) 7,200
 (3) 7,500 (4) 7,700
- 17.** A certain sum amounts to 5,832 in 2 years at 8% per annum compound interest, the sum is
 (1) 5,000 (2) 5,200
 (3) 5,280 (4) 5,400
- 18.** The compound interest on 6,000 at 10% per annum for $1\frac{1}{2}$ years, when the interest being compounded annually, is
 (1) 910 (2) 870
 (3) 930 (4) 900
- 19.** In what time 8,000 will amount to 9,261 at 10% per annum compound interest, when the interest is compounded half yearly?
 (1) $3\frac{1}{2}$ years (2) $1\frac{1}{2}$ years
 (3) $2\frac{1}{2}$ years (4) 2 years
- 20.** At what rate per cent per annum will a sum of 1,000 amounts to 1,102.50 in 2 years at compound interest?
 (1) 5% (2) 5.5%
 (3) 6% (4) 6.5%
- 21.** In how many years will a sum of 800 at 10% per annum compound interest, compounded semi-annually becomes 926.10 ?
 (1) $1\frac{1}{2}$ years (2) $1\frac{2}{3}$ years
 (3) $2\frac{1}{3}$ years (4) $2\frac{1}{2}$ years
- 22.** An amount of 6,000 lent at 5% per annum compound interest for 2 years will become
 (1) 600 (2) 6,600
 (3) 6,610 (4) 6,615
- 23.** In what time will 1000 amounts to 1331 at 20% per annum, compounded half yearly?
 (1) $1\frac{1}{2}$ years (2) 2 years
 (3) 1 year (4) $2\frac{1}{2}$ years

- 24.** The compound interest on 30,000 at 7% per annum for a certain time is 4,347. The time is
 (1) 3 years (2) 4 years
 (3) 2 years (4) 2.5 years
- 25.** A sum of 8000 will amount to 8820 in 2 years if the interest is calculated every year. The rate of compound interest is
 (1) 6% (2) 7%
 (3) 3% (4) 5%
- 26.** A principal of 10,000, after 2 years compounded annually, the rate of interest being 10% per annum during the first year and 12% per annum during the second year (in rupees) will amount to :
 (1) 12,000 (2) 12,320
 (3) 12,500 (4) 11,320
- 27.** The sum of money that yields a compound interest of 420 during the second year at 5% p.a is
 (1) 4,000 (2) 42,000
- (3) 8,000 (4) 21,000
- 28.** A man saves 2000 at the end of each year and invests the money at 5% compound interest. At the end of 3 years he will have :
 (1) 4305 (2) 6305
 (3) 4205 (4) 2205
- 29.** The time in which 80,000 amounts to 92,610 at 10% p.a. compound interest, interest being compounded semi annually is:
 (1) $1\frac{1}{2}$ years (2) 2 years
 (3) $2\frac{1}{2}$ years (4) 3 years
- 30.** A man borrows 21000 at 10% compound interest. How much he has to pay annually at the end of each year, to settle his loan in two years?
 (1) 12000 (2) 12100
 (3) 12200 (4) 12300

ANSWER KEY

1.	(2)	2.	(2)	3.	(3)	4.	(1)	5.	(3)	6.	(2)	7.	(3)	8.	(4)	9.	(3)	10.	(3)
11.	(1)	12.	(1)	13.	(3)	14.	(1)	15.	(4)	16.	(3)	17.	(1)	18.	(3)	19.	(2)	20.	(1)
21.	(1)	22.	(4)	23.	(1)	24.	(3)	25.	(4)	26.	(2)	27.	(3)	28.	(2)	29.	(1)	30.	(2)

PRACTICE SHEET-4

- 1.** If the compound interest on a certain sum for 2 years at 3% per annum is 101.50, then the simple interest on the same sum at the same rate and for the same time will be
 (1) 90.00 (2) 95.50
 (3) 100.00 (4) 98.25
- 2.** If the compound interest on a sum of money for 3 years at the rate of 5% per annum is 252.20, the simple interest on the same sum at the same rate and for the same time is
 (1) 220 (2) 240
 (3) 245 (4) 250
- 3.** On a certain sum of money the compound interest for 2 years is 282.15 and the simple interest for the same period of time is 270. The rate of interest per annum is
 (1) 6.07% (2) 10%
 (3) 9% (4) 12.15%
- 4.** If the compound interest on a sum for 2 years at $12\frac{1}{2}$ % per annum is 510, the simple interest on the same sum at the same rate for the same period of time is:
 (1) 400 (2) 480
 (3) 450 (4) 460
- 5.** The compound interest on a certain sum of money at a certain rate for 2 years is 40.80 and the simple interest on the same sum is 40 at the same rate and for the same time. The rate of interest is
 (1) 2% per annum (2) 3% per annum
 (3) 4% per annum (4) 5% per annum
- 6.** The compound interest on a certain sum of money invested for 2 years at 5% per annum is 328. The simple interest on the sum, at

- the same rate and for the same period will be
 (1) 320 (2) 308
 (3) 300 (4) 287
- 7.** Compound interest on a sum of money for 2 years at 4 per cent per annum is 2, 448. Simple interest of the same sum of money at the same rate of interest for 2 years will be
 (1) 2,500 (2) 2,400
 (3) 2,360 (4) 2,250
- 8.** At a certain rate per annum, the simple interest on a sum of money for one year is 260 and the compound interest on the same sum for two years is 540.80. The rate of interest per annum is
 (1) 4% (2) 6%
 (3) 8% (4) 10%
- 9.** The simple interest on a sum of money at 4% per annum for 2 years is 80. The compound interest in the same sum for the same period is
 (1) 82.60 (2) 82.20
 (3) 81.80 (4) 81.60
- 10.** The compound interest on a certain sum of money at 5% per annum for 2 years is 246. The simple interest on the same sum for 3 years at 6% per annum is
 (1) 435 (2) 450
 (3) 430 (4) 432
- 11.** The simple interest and compound interest (compounded annually) on a certain sum of money with a given rate for a period of 2 years are 900 and 954 respectively. The sum of money is
 (1) 3700 (2) 3650
 (3) 3850 (4) 3750
- 12.** The compound interest on a certain sum of money for 2 years at 10% per annum is 420. The simple interest on the same sum at the same rate and for the same time will be
 (1) 350 (2) 375
 (3) 380 (4) 400
- 13.** If the compound interest on a certain sum for 2 years at 4% p.a. is 102, the simple interest at the same rate of interest for two years would be
 (1) 200 (2) 50
 (3) 150 (4) 100
- 14.** There is 100% increase to an amount in 8 years, at simple interest. Find the compound interest of 8000 after 2 years at the same rate of interest.
 (1) 2500 (2) 2000
 (3) 2250 (4) 2125
- 15.** If the compound interest on a certain sum for two years at 12% per annum is 2,544, the simple interest on it at the same rate for 2 years will be
 (1) 2,400 (2) 2,500
 (3) 2,480 (4) 2,440
- 16.** The difference between compound and simple interest on a certain sum for 3 years at 5% per annum is Rs. 122. The sum is
 (1) 16,000 (2) 15,000
 (3) 12,000 (4) 10,000
- 17.** The difference between simple interest and compound interest of a certain sum of money at 20% per annum for 2 years is 48. Then the sum is
 (1) 1,000 (2) 1, 200
 (3) 1, 500 (4) 2, 000
- 18.** The difference between the compound interest and simple interest on 10,000 for 2 years is 25. The rate of interest per annum is
 (1) 5% (2) 7%
 (3) 10% (4) 12%
- 19.** If the difference between S.I. and C.I. for 2 years on a sum of money lent at 5% is 6, then the sum is
 (1) 2200 (2) 2400
 (3) 2600 (4) 2000
- 20.** On a certain sum of money lent out at 16% p.a. the difference between the compound interest for 1 year, payable half yearly, and the simple interest for 1 year is 56. The sum is
 (1) 1080 (2) 7805
 (3) 8750 (4) 5780
- 21.** If the amount is 2.25 times of the sum after 2 years at compound interest (compound annually), the rate of interest per annum is :
 (1) 25% (2) 30%
 (3) 45% (4) 50%

- 22.** A sum of money doubles itself in 4 years at compound interest. It will amount to 8 times itself at the same rate of interest in :
 (1) 18 years (2) 12 years
 (3) 16 years (4) 24 years
- 23.** A sum borrowed under compound interest doubles itself in 10 years. When will it become fourfold of itself at the same rate of interest?
 (1) 15 years (2) 20 years
 (3) 24 years (4) 40 years
- 24.** A sum of money becomes eight times of itself in 3 years at compound interest. The rate of interest per annum is
 (1) 100% (2) 80%
 (3) 20% (4) 10%
- 25.** A sum of money invested at compound interest doubles itself in 6 years. At the same rate of interest it will amount to eight times of itself in :
 (1) 15 years (2) 12 years
 (3) 18 years (4) 10 years
- 26.** A sum of money placed at compound interest doubles itself in 5 years. In how many years, it would amount to eight times of itself at the same rate of interest?
 (1) 10 years (2) 15 years
 (3) 7 years (4) 20 years
- 27.** A sum of money at compound interest doubles itself in 15 years. It will become eight times of itself in
 (1) 45 years (2) 48 years
 (3) 54 years (4) 60 years
- 28.** A sum of 12,000, deposited at compound interest becomes double after 5 years. How much will it be after 20 years?
 (1) 1,44,000 (2) 1,20,000
 (3) 1,50,000 (4) 1,92,000
- 29.** At what rate percent per annum of compound interest, will a sum of money become four times of itself in two years?
 (1) 100% (2) 75%
 (3) 50% (4) 20%
- 30.** A sum of money becomes double in 3 years at compound interest compounded annually. At the same rate, in how many years will it become four times of itself?
 (1) 4 years (2) 6 years
 (3) 6.4 years (4) 7.5 years

ANSWER KEY

1.	(3)	2.	(2)	3.	(3)	4.	(2)	5.	(3)	6.	(1)	7.	(2)	8.	(3)	9.	(4)	10.	(4)
11.	(4)	12.	(4)	13.	(4)	14.	(4)	15.	(1)	16.	(1)	17.	(2)	18.	(1)	19.	(2)	20.	(3)
21.	(4)	22.	(2)	23.	(2)	24.	(1)	25.	(3)	26.	(2)	27.	(1)	28.	(4)	29.	(1)	30.	(2)

PRACTICE SHEET-5

- 1.** A sum of money amounts to 4,840 in 2 years and to 5,324 in 3 years at compound interest compounded annually. The rate of interest per annum is :
 (1) 10% (2) 9%
 (3) 11% (4) 8%
- 2.** A certain sum of money amounts to 2,420 in 2 years and 2,662 in 3 years at some rate of compound interest, compounded annually. The rate of interest per annum is
 (1) 6% (2) 8%
 (3) 9% (4) 10%
- 3.** An amount of money at compound interest grows up to 3,840 in 4 years and up to 3,936 in 5 years. Find the rate of interest.
 (1) 2.5% (2) 2%
 (3) 3.5% (4) 2.05%
- 4.** A certain amount of money at $r\%$, compounded annually after two and three years becomes 1440 and 1728 respectively. r is
 (1) 5 (2) 10
 (3) 15 (4) 20

5. The compound interest on a certain sum for two successive years are 225 and 238.50. The rate of interest per annum is:
 (1) $7\frac{1}{2}\%$ (2) 5%
 (3) 10% (4) 6%
6. An amount of money appreciates to 7,000 after 4 years and to 10,000 after 8 years at a certain compound interest compounded annually. The initial amount of money was
 (1) 4,700 (2) 4,900
 (3) 4,100 (4) 4,300
7. A sum of money invested at compound interest amounts to 650 at the end of first year and 676 at the end of second year. The sum of money is:
 (1) 600 (2) 540
 (3) 625 (4) 560
8. A sum of money invested at compound interest amounts in 3 years to 2,400 and in 4 years to 2,520. The interest rate per annum is
 (1) 5% (2) 6%
 (3) 10% (4) 12%
9. A sum becomes 4500 after two years and 6750 after four years at compound interest. The sum is
 (1) 4000 (2) 2500
 (3) 3000 (4) 3050
10. A sum of money at compound interest will amount to 650 at the end of the first year and 676 at the end of the second year. The amount of money is
 (1) 1,300 (2) 650
 (3) 1,250 (4) 625
11. A builder borrows 2550 to be paid back with compound interest at the rate of 4% per annum by the end of 2 years in two equal yearly installments. How much will each installment be?
 (1) 1352 (2) 1377
 (3) 1275 (4) 1283
12. A man buys a scooter on making a cash down payment of 16224 and promises to pay two more yearly installments of equivalent amount in next two years. If the rate of interest is 4% per annum, compounded yearly, the cash value of the scooter, is
 (1) 40000 (2) 46824
 (3) 46000 (4) 50000
13. Kamal took 6800 as a loan which along with interest is to be repaid in two equal annual installments. If the rate of interest is $12\frac{1}{2}\%$, compounded annually, then the value of each installment is
 (1) 8100 (2) 4150
 (3) 4050 (4) 4000
14. A loan of 12,300 at 5% per annum compound interest, is to be repaid in two equal annual installments at the end of every year. Find the amount of each installment.
 (1) 6,651 (2) 6,615
 (3) 6,516 (4) 6,156
15. A sum of 210 was taken as a loan. This is to be paid back in two equal installments. If the rate of interest be 10% compounded annually, then the value of each installment is
 (1) 127 (2) 121
 (3) 210 (4) 225
16. Rs. 16,820 is divided between two brothers of age 27 years and 25 years. They invested their money at 5% per annum compound interest in such a way that both will receive equal money at the age of 40 years. The share (in Rs.) of elder brother is
 (1) 8,280 (2) 8,410
 (3) 8,820 (4) 8,000
17. A sum of money is paid back in two annual installments of Rs. 17, 640 each, allowing 5% compound interest compounded annually. The sum borrowed was
 (1) Rs. 32,800 (2) Rs. 32,200
 (3) Rs. 32,000 (4) Rs. 32,400
18. Mr. Dutta desired to deposit his retirement benefit of Rs. 3 lacs partly to a post office and partly to a bank at 10% and 6% interests respectively. If his monthly interest income was Rs. 2000, then the difference of his deposits in the post office and in the bank was
 (1) Rs. 50,000 (2) Rs. 40,000
 (3) Nil (4) Rs. 1,00,000

- 19.** The income of a company increases 20% per year. If the income is Rs. 26,64,000 in the year 2012, then its income in the year 2010 was :
 (1) Rs. 28,55,000 (2) Rs. 18,50,000
 (3) Rs. 28,20,000 (4) Rs. 21,20,000
- 20.** A person deposited a sum of 6,000 in a bank at 5% per annum simple interest. Another person deposited 5,000 at 8% per annum compound interest. After two years, the difference of their interests will be
 (1) 230 (2) 232
 (3) 832 (4) 600
- 21.** A money-lender borrows money at 4% per annum and pays the interest at the end of the year. He lends it at 6% per annum compound interest compounded half yearly and receives the interest at the end of the year. In this way, he gains 104.50 a year. The amount of money he borrows, is
 (1) 6,000 (2) 5,500
 (3) 5,000 (4) 4,500
- 22.** A sum of 13,360 was borrowed at $8\frac{3}{4}\%$ per annum compound interest and paid back in two years in two equal annual installments. What was the amount of each installment?
 (1) 5,769 (2) 7,569
 (3) 7,009 (4) 7,500
- 23.** Sita deposited 5,000 at 10% simple interest for 2 years. How much more money will Sita have in her account at the end of two years, if it is compounded semiannually.
 (1) 50 (2) 40
 (3) 77.50 (4) 85.50
- 24.** What does 250 amounts to in 2 years with compound interest at the rate of 4% in the 1st year and 8% in the second year?
 (1) 280 (2) 280.80
 (3) 468 (4) 290.80
- 25.** A man gave 50% of his savings of 84,100 to his wife and divided the remaining sum among his two sons A and B of 15 and 13 years of age respectively. He divided it in such a way that each of his sons, when they attain the age of 18 years, would receive the same amount at 5% compound interest per annum. The share of B was
 (1) 20,000 (2) 20,050
 (3) 22,000 (4) 22,050
- 26.** Find the rate percent per annum, if Rs. 2000 amounts to Rs. 2,315.25 in a year and a half, interest being compounded half yearly.
 (1) 11.5% (2) 10%
 (3) 5% (4) 20%
- 27.** A sum of money placed at compound interest doubles itself in 5 years. It will amount to eight times of itself at the same rate of interest in
 (1) 20 years (2) 10 years
 (3) 12 years (4) 15 years
- 28.** The sum of money which when given on compound interest at 18% per annum would fetch Rs. 960 more when the interest is payable half yearly than when it was payable annually for 2 years is :
 (1) Rs. 60,000 (2) Rs. 30,000
 (3) Rs. 40,000 (4) Rs. 50,000
- 29.** The amount on Rs. 25,000 in 2 years at annual compound interest, if the rates for the successive years be 4% and 5% per annum respectively is :
 (1) Rs. 30,000 (2) Rs. 26,800
 (3) Rs. 27,300 (4) Rs. 28,500
- 30.** The amount of Rs. 10,000 after 2 years, compounded annually with the rate of interest being 10% per annum during the first year and 12% per annum during the second year, would be (in rupees)
 (1) 11,320 (2) 12,000
 (3) 12,320 (4) 12,500

ANSWER KEY

1.	(1)	2.	(4)	3.	(1)	4.	(4)	5.	(4)	6.	(2)	7.	(3)	8.	(1)	9.	(3)	10.	(4)
11.	(1)	12.	(2)	13.	(3)	14.	(2)	15.	(2)	16.	(3)	17.	(1)	18.	(3)	19.	(2)	20.	(2)
21.	(3)	22.	(2)	23.	(3)	24.	(2)	25.	(1)	26.	(2)	27.	(4)	28.	(4)	29.	(3)	30.	(3)

CDS PYQ

1. The principal on whom a simple interest of Rs 55 will be obtained after 9 months at the rate of $3\frac{2}{3}\%$ per annum is.
[CDS 2013 (I)]
(a) Rs 1000 (b) Rs 1500
(c) Rs 2000 (d) Rs 2500
2. The difference between compound interest and simple interest for 2 yr at the rate of 10% over principal amount of X is Rs 10. What is the value of X?
[CDS 2014 (II)]
(a) Rs 100 (b) Rs 1000
(c) Rs 500 (d) Rs 5000
3. A sum of money becomes 3 times in 5 yr at simple interest. In how many years, will the same sum become 6 times at the same rate of simple interest?
[CDS 2014 (II)]
(a) 10 yr (b) 12 yr
(c) 12.5 yr (d) 10.5 yr
4. The difference between compound interest and simple interest at the same rate of interest R per cent per annum on Rs 15,000 for 2 years is Rs 96. What is the value of R?
[CDS 2015 (I)]
(a) 8
(b) 10
(c) 12
(d) Cannot be determined due to insufficient data
5. There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest on 12,000 after 3 years at the same rate of interest?
[CDS 2015 (I)]
(a) 2160 (b) 3120
(c) 3972 (d) 6240
6. In how much time would the simple interest on a principle be 0.125 time the principal amount at 10% per annum?
[CDS 2015 (I)]
- (a) $1\frac{1}{4}$ yrs (b) $1\frac{3}{4}$ yrs
(c) $2\frac{1}{4}$ yrs (d) $2\frac{3}{4}$ yrs
7. An automobile finance claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest at the rate of 10%, the effective rate of interest becomes
[CDS 2015 (I)]
(a) 10.25% (b) 10.5%
(c) 10.75% (d) 11%
8. A sum of Rs 10,000 is deposited for 1 year at the rate of interest 10% compounded half yearly. What will be the interest at the end of one year?
[CDS 2015 (II)]
(a) 1000 (b) 1025
(c) 1050 (d) 1100
9. If a sum of money at a certain rate of simple interest per year doubles in 5 years and at a different rate of simple interest per year becomes three times in 12 years, then the difference in the two rates of Simple interest per year is
[CDS 2016 (I)]
(a) 2% (b) 3%
(c) $3\frac{1}{3}\%$ (d) 4%
10. A sum of Rs 8,400 was taken as a loan. This is to be paid in two equal installments. If the rate of interest is 10% per annum, compounded annually, then the value of each installment is
[CDS 2017 (I)]
(a) 4,200
(b) 4,480
(c) 4,840
(d) None of the above
11. The difference between the simple and the compound interest on a certain sum of

money at 4% per annum in 2 years is Rs 10. What is the sum?

[CDS 2017 (I)]

- (a) 5,000 (b) 6,000
(c) 6,250 (d) 7,500

12. The difference between the compound interest (compounded annually) and simple interest on a sum of money deposited for 2 years at 5% per annum is Rs 15. What is the sum of money deposited?

[CDS 2017 (II)]

- (a) 6,000 (b) 4,800
(c) 3,600 (d) 2,400

13. A person borrowed Rs 5,000 at 5% rate of interest per annum and immediately lent it at 5.5%. After two years he collected the amount and settled his loan. What is the amount gained by him in this transaction?

[CDS 2018 (I)]

- (a) 25 (b) 50
(c) 100 (d) 200

14. A merchant commences with a certain capital and gains annually at the rate of 25%. At the end of 3 years he has 10,000. What is the original amount that the merchant invested?

[CDS 2018 (I)]

- (a) 5,120 (b) 5,210
(c) 5,350 (d) 5,500

15. The annual income of a person decreases by Rs 64 if the annual rate of interest decreases from 4% to 3.75%. What is his original annual income?

[CDS 2018 (I)]

- (a) 24,000 (b) 24,000
(c) 25,600 (d) 24,600

16. What the principle amount is with earns Rs 210 as compound interest for the second year at 5% per annum?

[CDS 2018 (II)]

- (a) Rs 2,000 (b) Rs 3,200
(c) Rs 4,000 (d) Rs 4,800

17. Two equal amounts were borrowed at 5% and 4% simple interest. The total interest after 4 years amounted to Rs 405. What was the total amount borrowed?

[CDS 2018 (II)]

- (a) Rs 1075 (b) Rs 1100
(c) Rs 1125 (d) Rs 1150

18. What is the least number of complete years in which a sum of money put at 40% annual compound interest will be more than treble?

[CDS 2019 (I)]

- (a) 3 (b) 4
(c) 5 (d) 6

19. A person divided a sum of Rs 17200 into three parts and invested at 5%, 6% and 9% per annum simple interest. At the end of two years, he got the same interest on each part of money. What is the money invested at 9%?

[CDS 2019 (I)]

- (a) Rs 3,200 (b) Rs 4,000
(c) Rs 4,800 (d) Rs 5,000

20. The rate of interest on two different schemes is the same and it is 20%. But in one of the schemes, the interest is compounded half yearly: and in the other the interest is, compounded annually. Equal amounts are invested in the schemes. If the difference of the returns after 2 years is Rs 482, then what is the. Principal amount in each scheme?

[CDS 2019 (II)]

- (a) 10,000 (b) 16,000
(c) 20,000 (d) 24,000

21. A lent 25000 to B and at the same time lent some amount to C at same 7% simple interest. After 4 years A received 11200 as interest from B and C. How much did A lend to C?

[CDS 2019 (II)]

- (a) Rs 20000 (b) Rs 25000
(c) Rs 15000 (d) Rs 10000

22. A sum was put at simple interest at certain rate of 2 years. Had it been put at 1 % higher rate of interest, it would have fetched Rs 24 more. What is the sum?

[CDS 2020 (I)]

- (a) Rs 500 (b) Rs 600
(c) Rs 800 (d) Rs 1200

23. The simple interest on a certain sum is one-fourth of the sum. If the number of years and the rate of annual interest are numerically equal, then the number of years is
[CDS 2020 (II)]
(a) 2.5 (b) 3
(c) 3.5 (d) 5
24. Sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched ₹ 500 more. What was the principal amount?
[CDS 2021 (I)]
(a) 2,000 (b) 1,800
(c) 1,600 (d) 1,200
25. The difference between the compound interest (compounded annually) and the of simple interest on a certain sum of money at 12% per annum for 2 years is ₹ 72. What is the principal amount?
[CDS 2021 (I)]
(a) 6,500 (b) 6,000
(c) 5,500 (d) 5,000
26. A sum of money compounded annually doubles itself in 5 years. In how many years will it become four times of itself?
[CDS 2021 (II)]
(a) 10 years (b) 12 years
(c) 15 years (d) 20 years
27. A person borrowed ₹ 9,000 at 7%, ₹ 12,000 at 8% and ₹ 15,000 at 9% simple interest per annum. He had to pay ₹ 50,700 at the end of n years. What is the value of n ?
[CDS 2022 (I)]
(a) 3 (b) 4
(c) 5 (d) 6
28. If C is the compound interest on ₹ 10,000 for one year at 4% per annum when compounded quarterly, then which one of the following is correct?
[CDS 2022 (II)]
(a) $C < ₹ 100$
(b) $₹ 100 < C < ₹ 200$
(c) $₹ 200 < C < ₹ 400$
(d) $C > ₹ 400$
29. What is the difference between simple interest and compound interest on ₹ 10,000 for two years at 20% per annum compounded half-yearly.
[CDS – 2023 (1)]
(a) ₹ 842 (b) ₹ 756
(c) ₹ 641 (d) ₹ 542
30. In a business dealing, A owes B ₹ 20,000 payable after 5 years, whereas B owes A ₹ 12,000 payable after 4 years. They want to settle it now at the rate of 5% simple interest. Who gives how much money in this settlement?
[CDS – 2023 (1)]
(a) both are at par
(b) B gives ₹ 6,000 to A
(c) A gives ₹ 6,000 to B
(d) A gives ₹ 4,000 to B
31. A sum of money at 20% rate of compound interest per annum becomes more than 100 times in n years. What is the least value of n ? (use $\log_{10} 2 = 0.301$, $\log_{10} 3 = 0.477$)
[CDS-2023 (2)]
(a) 23 (b) 24
(c) 25 (d) 26
32. A person borrowed ₹ 10,000 at 12% rate of interest per annum compounded quarterly for a period of 9 months. What is the interest paid by him to settle his account after 9 months?
[CDS-2023 (2)]
(a) ₹ 927.27 (b) ₹ 947.47
(c) ₹ 967.67 (d) ₹ 987.87
33. A sum of money invested at simple interest triples itself in 8 years and becomes n times in 20 years. What is the value of n ?
[CDS-2024 (1)]
(a) 5 (b) 6
(c) 7.5 (d) 9
34. Let x be the compound interest at the end of 3 years on a sum of ₹ 1000 at the rate of 10% compounded annually and y be the simple interest at the end of 3 years on a sum of ₹ 1000 at the annual rate of 11%. What is the difference between x and y ?
[CDS-2024 (1)]
(a) ₹ 16 (b) ₹ 15
(c) ₹ 5 (d) ₹ 1
35. If the rate of interest is 5%, then what would be the difference between compound

interest and simple interest received on Rs 10,000 (each) after 3 years from now?

[CDS-2024 (1)]

(a) Rs 175.25
(c) Rs 76.25

(b) Rs 152.25
(d) Rs 24.25

ANSWER KEY

1.	(c)	2.	(b)	3.	(c)	4.	(a)	5.	(c)	6.	(a)	7.	(a)	8.	(b)	9.	(c)	10.	(c)
11.	(c)	12.	(a)	13.	(b)	14.	(a)	15.	(c)	16.	(c)	17.	(c)	18.	(b)	19.	(b)	20.	(c)
21.	(c)	22.	(d)	23.	(d)	24.	(a)	25.	(d)	26.	(a)	27.	(c)	28.	(d)	29.	(c)	30.	(c)
31.	d	32.	(a)	33.	(b)	34.	(d)	35.	(c)										

CAPF PYQ

1. The difference between the compound interest and the simple interest for 2 years on a sum of money is Rs. 60. If the simple interest for 2 years is Rs. 1440, what is the rate of interest?

[CAPF 2017]

(a) $4\frac{1}{6}\%$

(b) $6\frac{1}{4}\%$

(c) 8%

(d) $8\frac{1}{3}\%$

2. A sum triples in ten years under compound interest at a certain rate of interest; the interest is being compounded annually. In how many years, it would become nine times

[CAPF 2020]

(a) 20 yrs
(c) 40 yrs

(b) 30 yrs
(d) 50 yrs

3. The difference of compound interest and simple interest of a sum of money at the rate of 5% per year for 2 years is Rs 250. The sum is

[CAPF 2021]

(a) Rs 100000
(c) Rs 40000

(b) Rs 80000
(d) Rs 120000

4. Suppose a bank gives an interest of 10% per annum compounded annually for a fixed deposit for a period of two years. What should be the simple interest rate per annum if the maturity amount after two years is to remain the same?

[CAPF 2022]

(a) 10%
(c) 11%

(b) 10.5%
(d) 12%

ANSWER KEY

1.	(d)	2.	(a)	3.	(a)	4.	(b)												
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